

Year-over-Year Analysis of Census HS-to-NAICS Mappings

Alternative Census Mapping Analysis

July 23, 2025

1 Executive Summary

This report analyzes changes in the Census Bureau’s HS-to-NAICS mapping data across multiple years, focusing on:

- Changes in HS code coverage
- Mapping consistency between years
- NAICS code evolution and restructuring
- Distribution patterns of additions and deletions

2 Analysis: 2021 to 2022

2.1 HS Code Changes

The following table summarizes the overall scope of HS code changes between 2021 and 2022, showing the total number of HS codes in each year and breaking down how many were added, deleted, or remained common across both years.

Metric	2021	2022	Change
Total HS Codes	19,266	19,758	+492
Added HS Codes	—	1,514	1,514
Deleted HS Codes	1,022	—	1,022
Common HS Codes		18,244	

Table 1: HS Code Changes from 2021 to 2022

2.2 Mapping Consistency

Mapping consistency measures whether HS codes that exist in both years maintain the same NAICS code assignment. High consistency (95%) suggests stable classification rules, while lower consistency indicates systematic reclassification efforts. Inconsistent mappings can reflect legitimate product reclassification, methodological updates, or data quality improvements.

Consistency Rate: 100.0%

- Consistent mappings: 18,241
- Inconsistent mappings: 3

2.3 NAICS Code Changes

This table shows changes in the NAICS classification system itself between 2021 and 2022. It tracks how many NAICS codes were added (new industry categories), deleted (discontinued categories), and the net change in the total number of industry classifications used in the mapping system.

Metric	2021	2022	Change
Total NAICS Codes	403	404	+1
Added NAICS Codes	—	1	1
Deleted NAICS Codes	0	—	0

Table 2: NAICS Code Changes from 2021 to 2022

2.4 Distribution Patterns

The distribution of HS code changes across NAICS codes reveals whether mapping updates are concentrated in specific industries or spread broadly across the economy. The concentration index (Herfindahl) measures this dispersion: values near 0 indicate widespread changes across many sectors, while higher values suggest changes concentrated in a few industries.

Change Distribution:

- Deleted HS codes spread across **142** NAICS codes
- Added HS codes spread across **147** NAICS codes
- NAICS codes with both additions and deletions: **135**
- Concentration Index - Deleted: **0.025**, Added: **0.021**

NAICS-Level Impact Analysis:

The table below shows all NAICS codes that experienced net changes in HS code assignments. Positive values indicate sectors that gained HS codes (expanding their trade coverage), while negative values show sectors that lost HS codes. This analysis reveals which industries are most affected by the mapping changes and helps identify sectors experiencing structural shifts in trade classification.

This table shows every NAICS industry that experienced a net change in HS code assignments. Each row represents an industry sector, with "Add" showing new HS codes assigned to that sector, "Del" showing codes removed, and "Net" showing the overall impact. Industries with positive net changes expanded their trade product coverage, while negative values indicate reduced coverage.

NAICS	Add	Del	Net
333517	60	19	+41
111219	48	9	+39
111419	32	0	+32
325199	84	58	+26
321211	50	24	+26
336120	38	16	+22
313210	38	16	+22
930000	34	16	+18
314994	23	8	+15
336612	33	20	+13
910000	22	11	+11
325411	17	6	+11
336411	44	33	+11
335129	22	13	+9
313230	8	0	+8
313220	11	3	+8
333318	10	2	+8
321999	24	16	+8
311224	20	12	+8
325180	14	6	+8
312111	8	1	+7
335121	12	6	+6
311411	6	0	+6
335122	12	6	+6
335312	19	13	+6
331110	0	6	-6
327999	5	11	-6
333999	16	11	+5
321213	12	7	+5
111411	7	2	+5
331491	10	5	+5
111422	7	2	+5
333994	8	3	+5
314110	11	6	+5
325412	8	4	+4
115114	4	0	+4
314120	7	11	-4
321911	8	4	+4

NAICS	Add	Del	Net
111211	4	0	+4
334310	11	7	+4
321918	8	4	+4
312130	4	0	+4
339910	8	4	+4
337211	5	1	+4
311911	9	6	+3
326299	5	2	+3
327211	3	0	+3
111334	0	3	-3
311514	5	2	+3
327212	9	6	+3
334419	5	2	+3
339950	6	3	+3
311511	1	4	-3
335991	4	1	+3
311710	4	2	+2
335911	2	4	-2
333120	7	5	+2
325613	17	15	+2
325414	9	11	-2
114111	0	2	-2
315220	73	71	+2
332993	2	0	+2
111140	13	11	+2
312230	9	7	+2
333249	8	6	+2
111335	4	6	-2
311421	4	2	+2
111333	0	2	-2
334220	12	10	+2
327215	6	4	+2
312140	2	0	+2
325211	5	3	+2
311611	2	0	+2
337215	26	24	+2
312120	2	0	+2

Table 3: All NAICS Codes with Net HS Code Changes (2021 to 2022) (showing 75 NAICS codes with —net change— ≥ 2)

2.5 New NAICS Code Analysis

This analysis examines newly introduced NAICS codes to understand the nature of classification system changes. New codes fall into three categories:

- **Consolidation:** Multiple existing NAICS codes merged into a single new code, typically reflecting industry convergence or administrative simplification.
- **Replacement:** One-to-one substitution where a new NAICS code directly replaces an existing code, often due to definitional updates or code restructuring.
- **Truly New:** Codes created for entirely new HS codes, representing emerging products or trade categories.

The Source NAICS Code(s) column shows which existing codes contributed to each new classification, providing insight into the structural evolution of the mapping system.

The following table lists every new NAICS code introduced in 2022, showing how many HS codes it covers, whether it represents a consolidation of multiple old codes, a direct replacement, or covers entirely new trade products, and which specific old NAICS codes contributed to its creation.

New NAICS Code	Category	HS Codes	Source NAICS Code(s)
115114	truly_new	4	N/A (new HS codes)

Table 4: New NAICS Code Classification Analysis

Key Observations: This period shows 0 consolidations, 0 replacements, and 1 truly new codes. Limited new code introduction suggests a stable classification period.

3 Analysis: 2022 to 2023

3.1 HS Code Changes

The following table summarizes the overall scope of HS code changes between 2022 and 2023, showing the total number of HS codes in each year and breaking down how many were added, deleted, or remained common across both years.

Metric	2022	2023	Change
Total HS Codes	19,758	19,879	+121
Added HS Codes	—	261	261
Deleted HS Codes	140	—	140
Common HS Codes		19,618	

Table 5: HS Code Changes from 2022 to 2023

3.2 Mapping Consistency

Mapping consistency measures whether HS codes that exist in both years maintain the same NAICS code assignment. High consistency (≥95%) suggests stable classification rules, while lower consistency indicates systematic reclassification efforts. Inconsistent mappings can reflect legitimate product reclassification, methodological updates, or data quality improvements.

Consistency Rate: 84.7%

- **Consistent mappings: 16,626**
- **Inconsistent mappings: 2,992**

3.3 NAICS Code Changes

This table shows changes in the NAICS classification system itself between 2022 and 2023. It tracks how many NAICS codes were added (new industry categories), deleted (discontinued categories), and the net change in the total number of industry classifications used in the mapping system.

Metric	2022	2023	Change
Total NAICS Codes	404	382	-22
Added NAICS Codes	—	21	21
Deleted NAICS Codes	43	—	43

Table 6: NAICS Code Changes from 2022 to 2023

3.4 Distribution Patterns

The distribution of HS code changes across NAICS codes reveals whether mapping updates are concentrated in specific industries or spread broadly across the economy. The concentration index (Herfindahl) measures this dispersion: values near 0 indicate widespread changes across many sectors, while higher values suggest changes concentrated in a few industries.

Change Distribution:

- Deleted HS codes spread across **42** NAICS codes
- Added HS codes spread across **29** NAICS codes
- NAICS codes with both additions and deletions: **19**
- Concentration Index - Deleted: **0.096**, Added: **0.091**

NAICS-Level Impact Analysis:

The table below shows all NAICS codes that experienced net changes in HS code assignments. Positive values indicate sectors that gained HS codes (expanding their trade coverage), while negative values show sectors that lost HS codes. This analysis reveals which industries are most affected by the mapping changes and helps identify sectors experiencing structural shifts in trade classification.

This table shows every NAICS industry that experienced a net change in HS code assignments. Each row represents an industry sector, with "Add" showing new HS codes assigned to that sector, "Del" showing codes removed, and "Net" showing the overall impact. Industries with positive net changes expanded their trade product coverage, while negative values indicate reduced coverage.

NAICS	Add	Del	Net
315250	26	0	+26
111422	26	1	+25
111219	60	38	+22
111332	15	3	+12
111339	13	2	+11
335910	10	0	+10
314120	11	3	+8
321211	10	2	+8
311421	7	0	+7
326122	6	0	+6
336110	6	0	+6
313210	0	6	-6
311611	0	6	-6
111419	6	11	-5
327991	5	1	+4
313230	0	4	-4

NAICS	Add	Del	Net
111120	8	4	+4
312111	0	3	-3
327211	0	3	-3
325199	6	3	+3
111130	7	4	+3
111998	4	1	+3
311221	6	3	+3
336111	0	3	-3
314994	0	2	-2
315220	0	2	-2
324110	2	0	+2
111320	2	0	+2
311211	2	0	+2
311411	0	2	-2
311224	6	4	+2

Table 7: All NAICS Codes with Net HS Code Changes (2022 to 2023) (showing 31 NAICS codes with —net change— ≥ 2)

3.5 New NAICS Code Analysis

This analysis examines newly introduced NAICS codes to understand the nature of classification system changes. New codes fall into three categories:

- **Consolidation:** Multiple existing NAICS codes merged into a single new code, typically reflecting industry convergence or administrative simplification.
- **Replacement:** One-to-one substitution where a new NAICS code directly replaces an existing code, often due to definitional updates or code restructuring.

- **Truly New:** Codes created for entirely new HS codes, representing emerging products or trade categories.

The Source NAICS Code(s) column shows which existing codes contributed to each new classification, providing insight into the structural evolution of the mapping system.

The following table lists every new NAICS code introduced in 2023, showing how many HS codes it covers, whether it represents a consolidation of multiple old codes, a direct replacement, or covers entirely new trade products, and which specific old NAICS codes contributed to its creation.

New NAICS Code	Category	HS Codes	Source NAICS Code(s)
335139	consolidation	65	335110, 335129
335910	consolidation	92	335911, 335912
337126	consolidation	46	337124, 337125
334610	consolidation	26	334613, 334614
333998	consolidation	71	333997, 333999
333248	consolidation	173	333244, 333249
322120	consolidation	132	322121, 322122
212115	replacement	4	212112
336110	consolidation	54	336111, 336112
321215	consolidation	17	321213, 321214
315120	consolidation	53	315110, 315190
212323	consolidation	13	212324, 212325
335131	replacement	12	335121
212290	consolidation	27	212291, 212299
316990	consolidation	102	316992, 316998
212390	consolidation	37	212391, 212392, 212393, +1 more
212114	consolidation	3	212111, 212113
212220	consolidation	10	212221, 212222
315250	consolidation	1907	315220, 315240, 315280
333310	consolidation	178	333314, 333316, 333318
335132	replacement	12	335122

Table 8: New NAICS Code Classification Analysis

Key Observations: This period shows 18 consolidations, 3 replacements, and 0 truly new codes. Consolidations dominate, suggesting administrative simplification efforts.

4 Analysis: 2023 to 2024

4.1 HS Code Changes

The following table summarizes the overall scope of HS code changes between 2023 and 2024, showing the total number of HS codes in each year and breaking down how many were added, deleted, or remained common across both years.

Metric	2023	2024	Change
Total HS Codes	19,879	19,978	+99
Added HS Codes	—	205	205
Deleted HS Codes	106	—	106
Common HS Codes		19,773	

Table 9: HS Code Changes from 2023 to 2024

4.2 Mapping Consistency

Mapping consistency measures whether HS codes that exist in both years maintain the same NAICS code assignment. High consistency (95%) suggests stable classification rules, while lower consistency indicates systematic reclassification efforts. Inconsistent mappings can reflect legitimate product reclassification, methodological updates, or data quality improvements.

Consistency Rate: 95.1%

- Consistent mappings: 18,810
- Inconsistent mappings: 963

4.3 NAICS Code Changes

This table shows changes in the NAICS classification system itself between 2023 and 2024. It tracks how many NAICS codes were added (new industry categories), deleted (discontinued categories), and the net change in the total number of industry classifications used in the mapping system.

Metric	2023	2024	Change
Total NAICS Codes	382	388	+6
Added NAICS Codes	—	9	9
Deleted NAICS Codes	3	—	3

Table 10: NAICS Code Changes from 2023 to 2024

4.4 Distribution Patterns

The distribution of HS code changes across NAICS codes reveals whether mapping updates are concentrated in specific industries or spread broadly across the economy. The concentration index (Herfindahl) measures this dispersion: values near 0 indicate widespread changes across many sectors, while higher values suggest changes concentrated in a few industries.

Change Distribution:

- Deleted HS codes spread across **22** NAICS codes
- Added HS codes spread across **21** NAICS codes
- NAICS codes with both additions and deletions: **13**
- Concentration Index - Deleted: **0.107**, Added: **0.133**

NAICS-Level Impact Analysis:

The table below shows all NAICS codes that experienced net changes in HS code assignments. Positive values indicate sectors that gained HS codes (expanding their trade coverage), while negative values show sectors that lost HS codes. This analysis reveals which industries are most affected by the mapping changes and helps identify sectors experiencing structural shifts in trade classification.

This table shows every NAICS industry that experienced a net change in HS code assignments. Each row represents an industry sector, with "Add" showing new HS codes assigned to that sector, "Del" showing codes removed, and "Net" showing the overall impact. Industries with positive net changes expanded their trade product coverage, while negative values indicate reduced coverage.

NAICS	Add	Del	Net	NAICS	Add	Del	Net
325412	40	16	+24	311411	2	0	+2
325199	37	13	+24	337122	2	0	+2
334413	19	0	+19	332119	2	0	+2
111219	41	22	+19	111310	2	1	+1
111419	18	5	+13	910000	4	3	+1
315250	0	12	-12	113210	0	1	-1
331110	8	0	+8	324110	0	1	-1
335910	0	5	-5	111320	0	1	-1
111333	8	4	+4	327991	0	1	-1
111422	0	4	-4	326199	1	0	+1
337127	4	0	+4	111339	6	5	+1
335999	0	2	-2	311211	0	1	-1
322120	2	4	-2	325414	2	1	+1
337121	2	0	+2	325180	2	1	+1

Table 11: All NAICS Codes with Net HS Code Changes (2023 to 2024)

4.5 New NAICS Code Analysis

This analysis examines newly introduced NAICS codes to understand the nature of classification system changes. New codes fall into three categories:

- **Consolidation:** Multiple existing NAICS codes merged into a single new code, typically reflecting industry convergence or administrative simplification.
- **Replacement:** One-to-one substitution where a new NAICS code directly replaces an existing code, often due to definitional updates or code restructuring.
- **Truly New:** Codes created for entirely new HS codes, representing emerging products or trade categories.

The Source NAICS Code(s) column shows which existing codes contributed to each new classification, providing insight into the structural evolution of the mapping system.

The following table lists every new NAICS code introduced in 2024, showing how many HS codes it covers, whether it represents a consolidation of multiple old codes, a direct replacement, or covers entirely new trade products, and which specific old NAICS codes contributed to its creation.

New NAICS Code	Category	HS Codes	Source NAICS Code(s)
327332	replacement	1	327999
331512	consolidation	2	331511, 331513
321912	replacement	6	113310
332721	consolidation	4	333111, 336350
326150	replacement	5	326113
115210	consolidation	5	11211X, 311613
332996	replacement	37	332919
326140	replacement	1	326113
1121XX	replacement	34	11211X

Table 12: New NAICS Code Classification Analysis

Key Observations: This period shows 3 consolidations, 6 replacements, and 0 truly new codes. Replacements are more common, indicating systematic code updates.

5 Summary and Insights

5.1 Key Findings

1. **Most Stable Period:** 2021-2022 showed high mapping consistency (100.0%) with broad, balanced changes across many sectors.
2. **Major Restructuring:** 2022-2023 experienced significant changes with 84.7% consistency and extensive NAICS consolidation.
3. **Focused Updates:** 2023-2024 showed targeted changes with 95.1% consistency, primarily in chemicals and electronics.
4. **Distribution Pattern:** Changes became increasingly concentrated over time, indicating more targeted updates rather than broad restructuring.

5.2 Concentration Trends

The concentration index (Herfindahl) shows how changes became more focused:

- **2021-2022:** Low concentration (0.025 deleted, 0.021 added) - widespread changes
- **2022-2023:** Medium concentration (0.096 deleted, 0.091 added) - consolidation period
- **2023-2024:** High concentration (0.107 deleted, 0.133 added) - targeted updates

5.3 Methodology Notes

- Data source: Census Bureau HTS import mapping files (imp-code_*.txt)
- Analysis covers years: 2021, 2022, 2023, 2024
- Concentration measured using Herfindahl index
- Categories: truly_new (all new HS codes), replacement (1-to-1 substitution), consolidation (many-to-1 merger)